

E-06-alignment-persistence v1: can dynamics sustain omega in compressive eigenspace of S?

ZFL Lab

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MISSION: Audit whether Navier-Stokes dynamics can sustain vorticity in compressive eigenspaces of S , relevant to W_- bottleneck. INPUT E-05: $\omega^T S \omega = \sum \lambda_i \omega_i^2$, $\sum \lambda_i = 0$, $\frac{1}{2} D_t |\omega|^2 = \omega^T S \omega + \frac{\nu}{2} \Delta |\omega|^2 - \nu |\nabla \omega|^2$. $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$.

FORBIDDEN: CF alignment $\Rightarrow \xi \approx e_1$. Instantaneous alignment $\Rightarrow$ persistent alignment. η_- small $\Rightarrow W_- \in L_t^1$ without quantitative estimate. Promotion to C_E^{strong} .

AUDIT:

1. Instantaneous geometry:

Where $\omega \neq 0$, define $\xi = \omega/|\omega|$, $\omega_i = \omega \cdot e_i(S)$, $\lambda_1 \geq \lambda_2 \geq \lambda_3$, $\sum \lambda_i = 0$.

$$\eta_i = \frac{\omega_i^2}{|\omega|^2} \geq 0, \quad \sum \eta_i = 1, \quad \frac{\omega^T S \omega}{|\omega|^2} = \sum \lambda_i \eta_i.$$

Define compressive fraction:

$$\boxed{\eta_-(x, t) = \sum_{\lambda_i(x, t) < 0} \eta_i(x, t)}, \quad \eta_+(x, t) = 1 - \eta_-.$$

Then

$$W(t) = \int |\omega|^2 \sum \lambda_i \eta_i dx, \quad W_-(t) = \int |\omega|^2 (\sum \lambda_i \eta_i)_- dx.$$

If $\lambda_3 < 0$ dominant, $(\sum \lambda_i \eta_i)_- \geq |\lambda_3| \eta_-$ modulo $\lambda_{1,2}$ cancellations. So

$$\int \eta_- |\omega|^2 dx \text{ large} \Rightarrow W_- \text{ potentially large, unless } |\lambda_3| \text{ small.}$$

No bound on η_- from $\text{tr } S = 0$ alone. Spectral constraint only says at least one $\lambda_i \geq 0$, not that η_- small.

2. Dynamic persistence:

Vorticity equation: $D_t \omega = S \omega + \Omega \omega + \nu \Delta \omega$, Ω skew (rotation, preserves $|\omega|$). For ξ , formally where $\omega \neq 0$,

$$D_t \xi = S \xi - (\xi^T S \xi) \xi + \Omega \xi + \nu \Delta \xi + 2\nu |\nabla \xi|^2 \xi + \text{lower terms from } \nabla |\omega|.$$

Exact form: projection of $S \xi$ onto tangent of S^2 : $P_{\xi^\perp} S \xi$. So S drives ξ toward e_1 (max λ) via $S \xi - (\xi^T S \xi) \xi$.

Thus mechanism: stretching tends to align ξ with e_1 , which would make $\eta_- \rightarrow 0$. But Ω , transport, and diffusion $\nu \Delta \xi$ can counteract.

Key question: can dynamics maintain $\xi \approx e_3$ for significant time interval near T_* ?

$$\text{Persistence question: } \int_{T_*-\delta}^{T_*} \int \eta_- |\omega|^2 dx dt \stackrel{?}{\text{can be large}}$$

No result in PROVEN gives answer. CF requires $\nabla \xi$ bounded to prevent blow-up, not that $\xi \rightarrow e_1$. DHY requires bounded curvature of vortex lines, not alignment.

3. Two target structures:

Structure A (control W_- by small η_-):

$$\eta_- \leq \delta \Rightarrow W_-(t) \leq C(\delta) \int |\lambda_3|_- |\omega|^2.$$

If $|\lambda_3|$ controlled by $\|\nabla u\|$, still no L_t^1 . Needs $|\lambda_3| \in L_t^1$ weighted by $|\omega|^2$, not given.

But if $\eta_- \leq \delta$ pointwise with small δ , then $\sum \lambda_i \eta_i \geq (1-\delta)\lambda_1 + \delta\lambda_3$. Since $\lambda_1 = -(\lambda_2 + \lambda_3)$, not automatically ≥ 0 unless λ_2 small. So even η_- small does not guarantee $W_- \in L^1$ without eigenvalue magnitude control.

Structure B (compressive persistence restricts growth): From enstrophy identity:

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 = W_+ - W_- - \nu \|\nabla \omega\|_2^2? \text{ Wait sign: } W - \nu \|\nabla \omega\|_2^2.$$

Actually $W = W_+ - W_-$, so $\frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 = W_+ - W_- - \nu \|\nabla \omega\|_2^2$? No, correct: $\frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 = W - \nu \|\nabla \omega\|_2^2 = W_+ - W_- - \nu \|\nabla \omega\|_2^2$.

If dynamics forces ω in compressive region, then W_+ small, so $\frac{d}{dt} \|\omega\|_2^2 \leq -2W_- - 2\nu \|\nabla \omega\|_2^2 \leq 0$, i.e., enstrophy decreasing. But C_E^{weak} requires $\int W_+ = \infty$, which implies enstrophy must grow in some sense? Actually from C_E^{weak} we have $\limsup \int_0^t W = +\infty$, which with $W = \frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 + \nu \|\nabla \omega\|_2^2$ implies $\|\omega\|_2^2 + \int \nu \|\nabla \omega\|^2$ unbounded? Let's check: $\int_0^t W = \frac{1}{2} \|\omega(t)\|_2^2 - \frac{1}{2} \|\omega(0)\|_2^2 + \nu \int_0^t \|\nabla \omega\|_2^2$. So $\int W_+ = \infty$ does not directly force growth if W_- large, but $\int W = \infty$ from C_E^{weak} does force $\frac{1}{2} \|\omega\|_2^2 + \nu \int \|\nabla \omega\|_2^2 \rightarrow \infty$.

Thus: if persistent compressive alignment makes $W \leq 0$, then $\frac{d}{dt} \|\omega\|_2^2 \leq 0$ modulo diffusion? Actually $W \leq 0$ implies enstrophy decreasing if $\nu > 0$. That would contradict $\int W = \infty$ which requires overall positive W average diverging.

Hence we get potential incompatibility:

$$\text{Persistent } \eta_- \approx 1 \Rightarrow W \leq 0 \text{ on average} \Rightarrow \int W \text{ bounded} \Rightarrow \text{contradiction with } C_E^{weak}$$

But this is not rigorous without quantifying λ_i magnitudes: if ω sits where $|\lambda_3|$ small, W could still be small but not negative large.

So Structure B yields:

$$\text{Compressive persistence} \Rightarrow \text{restriction on growth of } |\omega|, \text{ not yet } W_- \in L_t^1$$

4. Forbidden jumps avoided:

No CF $\Rightarrow \xi \approx e_1$. CF gives regularity of ξ , not alignment. No instantaneous η_- small $\Rightarrow$ persistent small. No η_- small $\Rightarrow W_- \in L_t^1$ without temporal bound on λ .

OUTPUT:

$$\eta_- = \sum_{\lambda_i < 0} \omega_i^2 / |\omega|^2 \text{ exact diagnostic of compressive fraction}$$

$$\frac{\omega^T S \omega}{|\omega|^2} = \sum \lambda_i \eta_i \text{ exact}$$

$D_t \xi$ driven by $P_{\xi^\perp} S \xi + \Omega \xi + \nu \Delta \xi$ — alignment with e_1 is tendency, not guarantee

Persistent compressive alignment $\Rightarrow \frac{d}{dt} \|\omega\|_2^2$ suppressed, tension with $C_E^{weak} : \int_0^{T_*} W = \infty$

No proof that dynamics can sustain $\eta_- \approx 1$ near T_* — OPEN/BLOCKED

Neither Structure A nor B gives $W_- \in L_t^1$ without extra quantitative control of $\lambda_i(t)$

$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ intact. No new C added. G remains locked at 0a37811.

Next experiment: quantitative estimate for $D_t \eta_-$ using $S, \Omega, \nu \Delta$ terms, to test if diffusion destroys compressive persistence.